Statistical Modeling — Air Quality Analysis

Air Quality
in Peace & War

Parma, Italy · vs · Kyiv, Ukraine (Oct 2023)

3,600	**818.85**	**54×**	**R^2=0.93**
Samples per city	Kyiv PM2.5 peak (μg/m³)	Above WHO daily limit	Kyiv regression fit

Clustering & Regression Analysis · MATLAB

Presentation Agenda

01

Project Motivation

- Why air quality in conflict zones?
- The Gaza problem: no data as evidence = a message
- Why we chose Kyiv as proxy

02

Data & Sensor Setup

- Raspberry Pi + SDS011 sensor
- Parma: parks + bar area collection
- Kyiv: SaveEcoBot station #1651

03

Statistical Analysis(MATLAB)

- Descriptive statistics & WHO limits
- K-Means clustering (K=3)
- Linear regression PM10 ~ PM2.5

04

Insights & Conclusions

- War creates unmistakable air signatures
- Parma's unexpected PM10 problem
- What the data tells us

The Idea Started in Gaza

Why we cared — and what the absence of data actually means

The Original Research Goal

We wanted to measure what people in Gaza breathe during the 2023–2024 war.

PM2.5 and PM10 from explosions, burning buildings, and displaced dust are acutely dangerous.

We searched every major open-data platform: WHO, IQAir, OpenAQ, SaveEcoBot, AQICN.

Result: zero ground-sensor PM2.5/PM10 records from Gaza during the war.

The conflict destroyed every monitoring station. This is not a data gap — it is a consequence of the destruction itself.

Confirmed: Abulibdeh et al. (2025, Global Environmental Change) found the same absence of ground data and was also forced to rely on satellite measurements instead.

Our Solution

We pivoted to Kyiv, Ukraine — a city currently under war with real open-access ground sensor data matching our SDS011 hardware.

"Gaza was the idea. Kyiv was the data."

A Finding in Itself

The absence of Gaza data is not a research failure. It is evidence that the conflict erased the infrastructure needed to document its own environmental impact.

No data = The data.

Two Datasets — Same Sensor, Different Worlds

Dataset 1 — Parma, Italy

Source: Raspberry Pi + SDS011

Location: Parma parks + open-air bar

Context: Normal urban life — fresh air and cigarette smoke

Samples: 3,600 paired PM2.5 / PM10 readings

5.30 PM2.5 Mean µg/m³	**26.2** PM2.5 Peak µg/m³
47.0 PM10 Mean µg/m³	**4.7%** Exceed WHO PM25

Note: PM10 often exceeds WHO — road dust, traffic, bar area smoke contribute to coarse particles even in peaceful Parma.

Dataset 2 — Kyiv, Ukraine

Source: SaveEcoBot station #1651 (real sensor data)

Event: Missile attack — October 27, 2023

Window: Oct 25–31, 2023 (3,600 consecutive readings)

Original file: 2.8M rows (2019–2026) → extracted worst window

10.61 PM2.5 Mean µg/m³	**818.85** PM2.5 Peak µg/m³
23.4 PM10 Mean µg/m³	**17.2%** Exceed WHO PM25

The high PM2.5 std dev for Kyiv (33.73) vs Parma (4.46) tells the real story: war creates extreme unpredictability, not just higher averages.

The Code: Four Steps, One Script

Load & Normalise

load() reads both .mat files. zscore() normalises so PM10 scale does not overpower PM2.5 before clustering.

Elbow Method

kmeans() runs K=1 to 6. We plot total inertia and look for the bend — the point where adding more clusters stops helping.

K-Means K=3

Final clustering with 30 replicates. grpstats() verifies centroids. gscatter() plots clusters with two panels: full view + zoomed view.

Regression (Parma)

fitlm() fits PM10 = intercept + slope × PM2.5 for Parma only. predict() draws the fitted line and 95% confidence band.

Key MATLAB Functions

`zscore(X)`

Normalise — same as (X-mean)/std

`kmeans(X,k,'Replicates',30)`

Run k-means 30 times, keep best result

`grpstats(X,idx)`

Compute group means to verify centroids

`assert(...)`

Check the verification — crash if wrong

`gscatter(x,y,group)`

Scatter plot with colour per group

`fitlm(x,y)`

Fit linear model: slope, intercept, R^2, p

`coefCI(mdl,0.05)`

95% confidence interval on the slope

`predict(mdl,xfit)`

Generate fitted line + confidence band

Visualising the Raw Data

What do 3,600 sensor readings look like for each city?

Parma: Stable

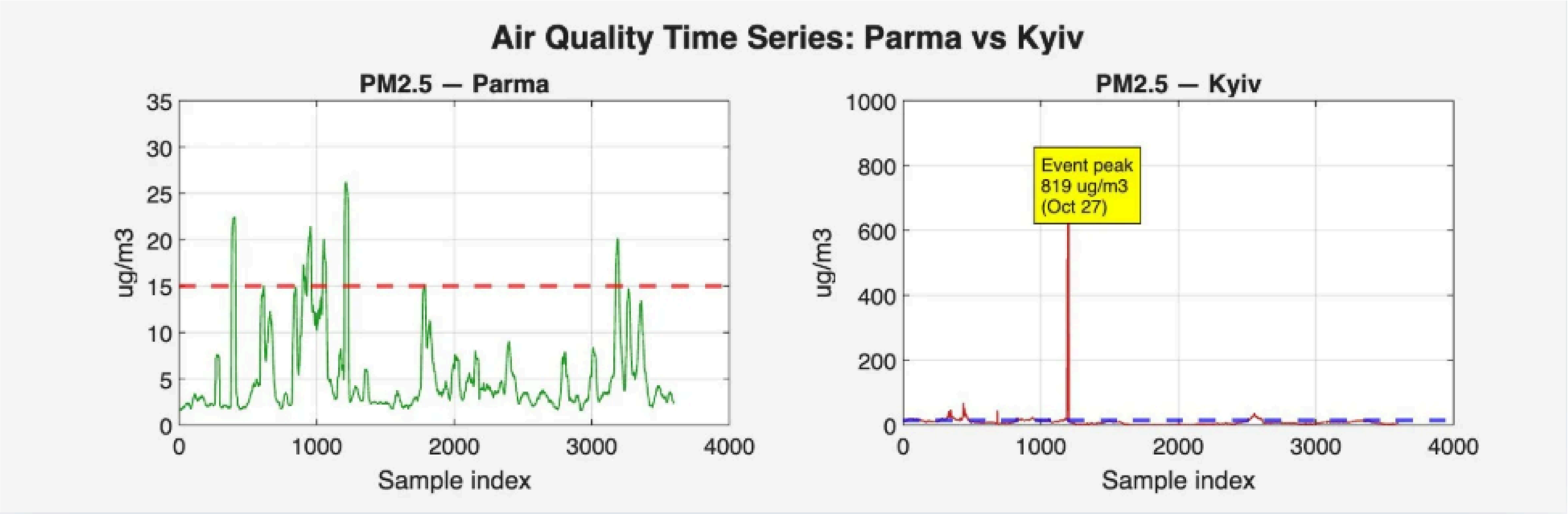
Mostly below WHO limit of 15 µg/m³. Small spikes from bar area smoke and traffic. Max = 26 µg/m³.

Kyiv: One Event

One catastrophic spike at sample ~1,200. Before and after it: relatively normal air. The event dominates everything.

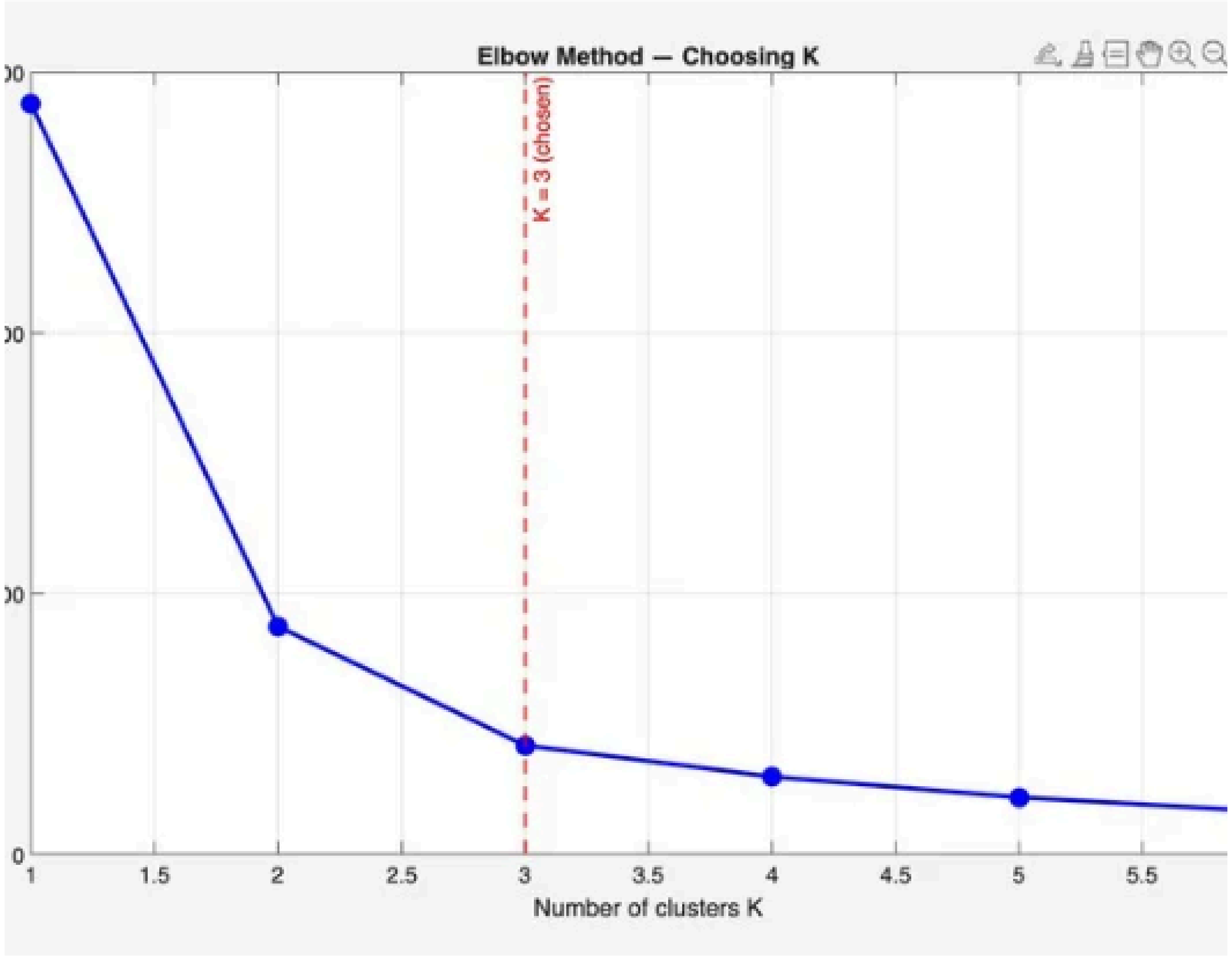
Why This Matters

These charts show WHY we need clustering — the eye can see something unusual happened. Now we let the algorithm confirm it.



Elbow Method: How Many Clusters?

We test K=1 to 6 and look for where the curve stops dropping sharply



What is inertia?

The total distance of all points from their cluster centre. Lower = tighter, better-defined clusters.

Why K=3?

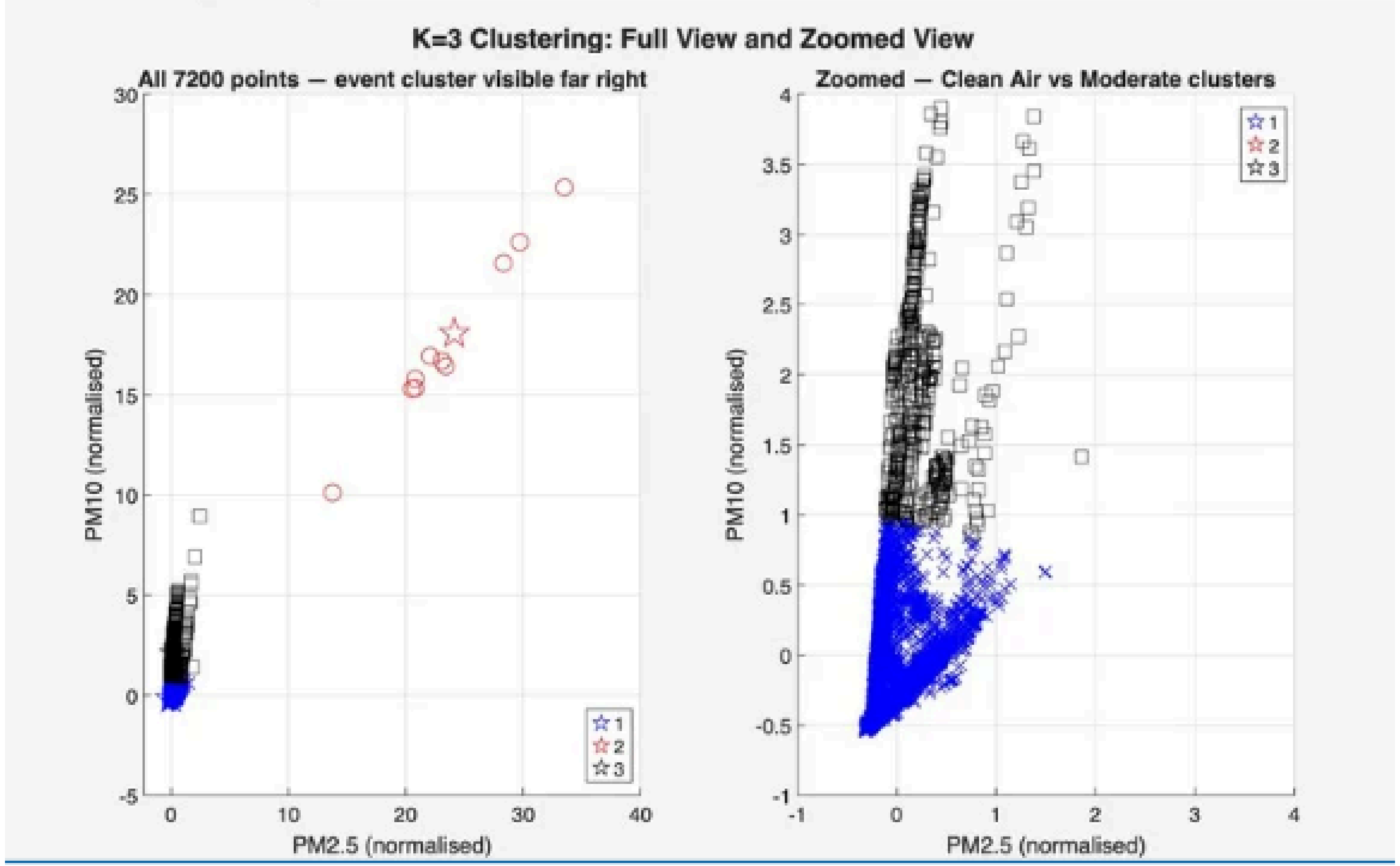
K=1→2: huge drop. K=2→3: still a big drop. K=3→4: small drop. The curve elbows at K=3 — adding more clusters stops helping significantly.

What K=3 means

Three natural air quality regimes exist in the data: Clean Air, Moderate, and Extreme. The algorithm found this without being told.

The Algorithm Found the Event — On Its Own

7,200 data points, no labels, no context about what happened

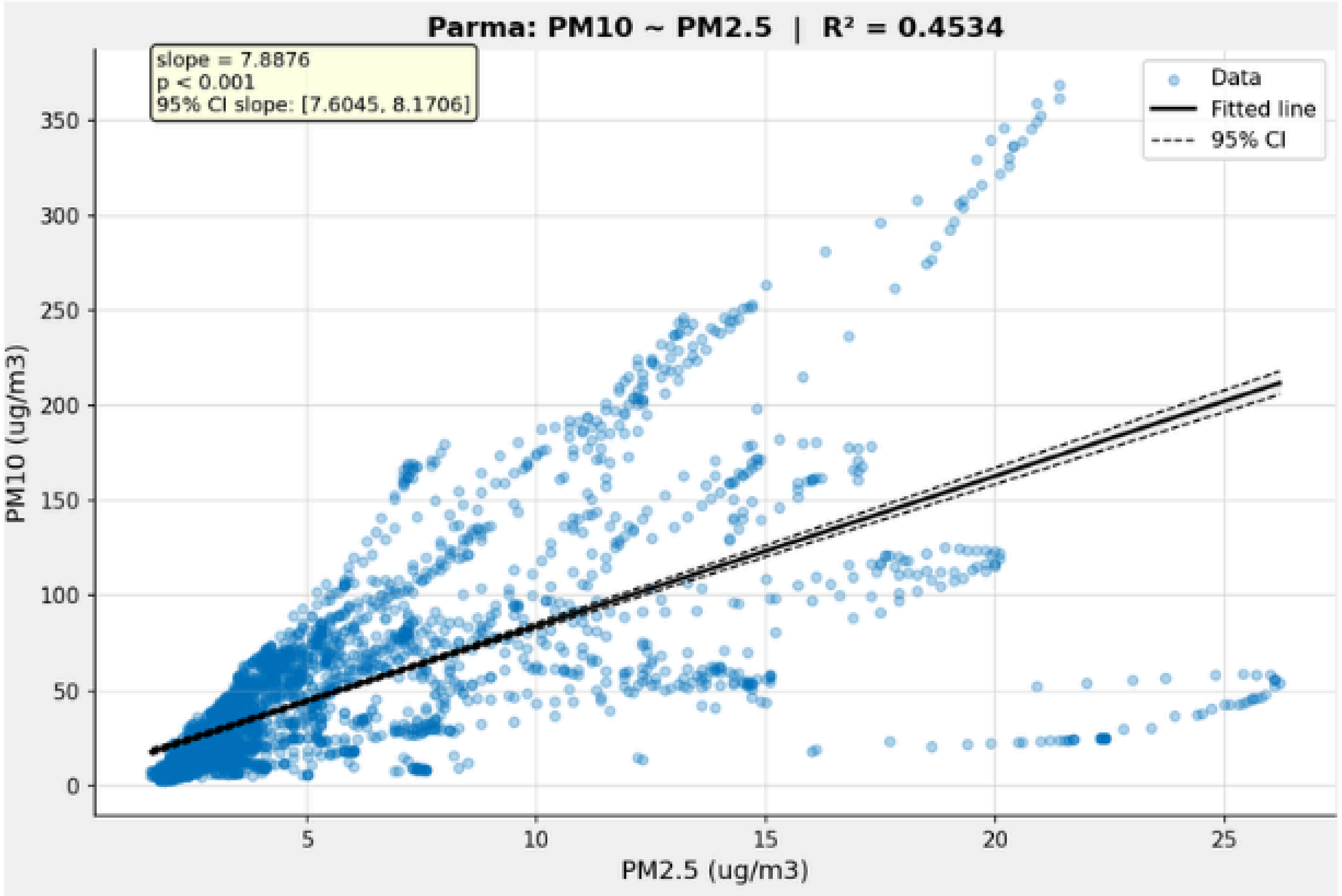


Left panel: all 7,200 points — the red circles far to the right are the 11 extreme readings from Oct 27. Completely isolated from everything else.

Right panel: zoomed into the normal range — now you can see the blue (Clean Air) and black (Moderate) clusters clearly separated.

Does PM2.5 Predict PM10 in Normal Urban Life?

fitlm() fits PM10 = intercept + slope × PM2.5 for Parma only



Why Parma only?

Kyiv has one extreme event that pulls the entire regression line. The R^2 would be near-perfect (0.93) but it just reflects that single spike — not a real PM2.5/PM10 relationship.

Parma has natural day-to-day variation from multiple sources, so the regression tells something genuinely meaningful.

$R^2 = 0.4534$

45% of PM10 variation is explained by PM2.5

slope = 7.89

Each +1 μg/m³ PM2.5 → +7.89 μg/m³ PM10

$p < 0.001$

Slope is statistically significant

Four Things the Data Tells Us

The Algorithm Found the Event Alone

K-Means received 7,200 unlabelled rows of two numbers. It returned Cluster 3: 11 points, all from Oct 27, PM2.5 centroid = 593 μg/m³. Zero Parma readings in this cluster. No dates, no city names, no context provided.

Two Views Tell the Complete Story

Full view: the extreme cluster sits completely isolated, showing how severe the separation is. Zoomed view: Clean Air and Moderate clusters have real structure — not everything in the data is extreme.

Peacetime Parma Is Not Automatically Clean

35.8% of Parma's PM10 readings exceeded the WHO limit. Road dust, traffic, and the bar area smoke create chronic coarse-particle exposure that is easy to overlook next to dramatic acute events.

Real Sensor Data, Real Results

Our Raspberry Pi produced data precise enough for K-Means to isolate an anomaly and regression to find a meaningful PM2.5/PM10 relationship. The hardware provided for this project delivered scientifically valid output.

Conclusions & References

Our original hypothesis was that war would produce the worst air quality in both pollutants. What we discovered was more nuanced: the acute catastrophic PM2.5 spike in Kyiv is unmistakable but Parma's PM10 exceeded WHO limits 35.8% of the time. The reason is our collection site: a park near an open-air bar with active smokers. This shows that peacetime urban chronic exposure is a real, ongoing health concern separate from war's acute events. Both matter, but in fundamentally different ways.

Future Directions

Add temperature + humidity · real-time threshold alerts · test on additional locations (Gaza via satellite)

Open-source ground sensor data (SaveEcoBot) is compatible with our Raspberry Pi hardware, enabling academic comparison.

References

[1] SaveEcoBot Platform

Kyiv sensor network, station #1651
saveecobot.com — CC BY 4.0

[2] MDPI Cities (2025)

Katrenko et al. — Wartime Air Pollution Kyiv
Cities 9(11):477 · doi:10.3390/cities9110477

[3] WHO Guidelines 2021

PM2.5: 15 μg/m³ | PM10: 45 μg/m³
(24-hr mean)

[4] MathWorks MATLAB

kmeans(), fitlm(), grpstats(), gscatter()
Statistics & ML Toolbox

[5] Abulibdeh et al. (2025)

Air pollution in Gaza post-Oct 7 · Global Environmental Change, 91, 102975

Thank You!

Questions & Discussion Welcome

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